

HUMAN COGNITIVE COMPATIBILITY AND COLLABORATION IN THE DECISION-MAKING PROCESS

DOI:10.36962/ECS108/4-6/2026-64

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RESUME

Human daily activity is continuously associated with decision-making, which may range from simple to complex and strategic, such as life planning, allocating financial investments, and making decisions at various levels within the workplace. The effectiveness of decision-making depends not only on the availability of information, but also on an individual's ability to perceive, process, and evaluate it. In this process, one of the key factors is **the identification of an individual's cognitive style and the recognition of compatibility between different cognitive styles**.

Cognitive compatibility refers to the alignment between an individual's mental capacities and the information they receive. It encompasses processes such as perception, memory, attention, intuition, and analytical ability. When information aligns with a person's cognitive style, decision-making tends to be faster, more accurate, and requires less effort. Conversely, when information is inconsistent or difficult to comprehend, the risk of errors and poor decision-making increases.

The individual capabilities of a decision-maker are particularly important when decisions are made under conditions of crisis, stress, and informational uncertainty. In such situations, decisive importance is attached not only to professional knowledge and experience, but also to an individual's psychological characteristics, intellectual potential, motivations, values, and internal attitudes.

This article presents and emphasizes, for the purpose of achieving optimal decision-making, the cognitive compatibility styles of individuals involved in the decision-making process, their key characteristics, and cognitive compatibility matrices, both in stable conditions and in dynamic or crisis situations.

The article formulates the **"golden rule" of temperaments across different styles and presents cognitive compatibility matrices**.

Keywords: decision-making, cognitive compatibility, cognitive balance, crisis, collaboration, "golden rule," cognitive compatibility matrix, decision-making effectiveness, emotional state.

MAIN TEXT

Human daily life continuously requires decision-making that varies in complexity, strategic nature, and significance. These decisions may relate to routine tasks, such as time management and the allocation of financial resources, as well as to more complex strategic issues, such as career choice, investment planning, medical treatment, the formulation of public policy, or the development of business strategies. The effectiveness of decision-making depends not only on the availability of information, but also on an individual's ability to perceive, analyze, evaluate, and purposefully apply that information. [1]

In this process, a particularly important role is assigned **to human cognitive systems and their compatibility with the environment and the presented information**. Cognitive compatibility refers

to the harmony between an individual's mental capacities and the information received, which ensures the optimization of the decision-making process. When decision-making takes place in a cognitively compatible environment, the process becomes faster, more accurate, less exhausting, and less stressful.

The decision-making process involves several cognitive stages: perception of information, processing, generation and evaluation of alternatives, making a final choice, and implementing the decision. These processes involve not only rational thinking, but also **emotions, intuition, experience, and environmental perception**. Cognitive compatibility ensures the harmony and balance of these processes.

For example, for visually oriented individuals, decision-making based on graphs, diagrams, or visualized data becomes faster and more accurate, as this format aligns with their perceptual style. On the other hand, for text-oriented individuals, the same data is more easily understood in a written format, which accelerates information processing and improves the quality of decisions.

It is important to highlight the **“four classical temperament theory,”** which originates from the theory of Hippocrates. Temperamental compatibility refers not only to the “matching of personalities,” but also to how well individuals align in terms of their information processing speed, emotional responsiveness, risk perception, conflict management style, and decision-making model. [2]

The four main cognitive styles are generally distinguished:

1. **Sanguine** – typically characterized by activity and sociability.
2. **Choleric** – goal-oriented, energetic, and distinguished by leadership qualities.
3. **Melancholic** – a sensitive, responsible individual oriented toward details.
4. **Phlegmatic** – a calm, balanced, and patient individual.

Cognitive compatibility between individuals refers to the alignment of **thinking styles, information processing speed, risk attitudes, and emotional responses**. When members of a group share similar cognitive structures, the decision-making process becomes more efficient. If one member is characterized by an analytical, detail-oriented approach, while another has a strategic and long-term perspective, their collaboration leads to higher-quality decisions. However, when a group is dominated by a single type of thinking (for example, a tendency toward overly risky decisions), the likelihood of excessive risk increases. For instance, in a technology startup, one co-founder may be innovation-oriented and willing to take significant risks, while the other prioritizes financial stability. If properly managed, their cognitive differences create balance and reduce the likelihood of extreme decision-making. [3]

In the decision-making process, it is important to take into account the **cognitive load** of the individuals involved, which refers to the mental effort required during information processing and represents a significant factor. Under conditions of high cognitive load, individuals are often unable to fully comprehend information, attention becomes overloaded, and memory capacity is strained; consequently, the quality of decisions decreases. **Cognitive compatibility** reduces this load, as information is structured, easily perceivable, and adapted to the individual's capabilities. For example, complex technical instructions that do not correspond to the user's level of knowledge lead to confusion and incorrect decisions, whereas simple and logically organized information improves analysis, increases the likelihood of correct decision-making, and supports more efficient use of memory. [4]

Individuals differ from one another in **cognitive abilities, experience, education, and thinking styles**. Therefore, the same information may be differently compatible for different individuals. **Cognitive compatibility** is dynamic and should be assessed based on individual factors. In Table 1, we present a comparative characterization of temperaments in the context of decision-making.

REGARDING THEIR COMPATIBILITY IN THE WORKPLACE:

Choleric + Melancholic (strategic pair): This combination demonstrates high efficiency, combining strategy and analysis. A potential risk is the choleric's dominance or pressure and the melancholic's emotional sensitivity. An ideal working environment for this pair is a structured organization.

Sanguine + Phlegmatic (social pair): The sanguine individual accelerates processes, while the phlegmatic balances emotional fluctuations. The sanguine may sometimes experience boredom in the workflow. The strength of their collaboration lies in social harmony. The outcome is a positive working environment and long-term stability.

Table. Characteristics of Temperaments

Characteristic	Choleric	Sanguine	Melancholic	Phlegmatic
Decision style	Fast, dominant	Impulsive, intuitive	Analytical, calculated	Stable, balanced
Main motivator	Result and control	Recognition and enthusiasm	Accuracy and logic	Harmony and security
Need for information	Minimal (just the essence)	General (big picture)	Maximum (details)	Moderate (methodology)
Reaction to stress	Aggression / Action	Emotionality / Optimism	Isolation / Verification	Passivity / calmness
Risk propensity	High	Very high	Low	Minimum

Source: compiled by the author

Choleric + Sanguine (extraverted energy): Both are oriented toward speed. The main risk is the neglect of details and the production of superficial decisions. The most optimal working environment for this combination is startups and dynamic business settings.

Melancholic + Phlegmatic (introverted stability): Both are oriented toward quality and process. The main risk is excessive delay in decision-making.

Sanguine + Melancholic: This represents an emotional and analytical combination. Specifically, the sanguine generates ideas, while the melancholic verifies and refines details. However, the sanguine often perceives the melancholic as overly critical.

Choleric + Phlegmatic: The choleric takes the lead, while the phlegmatic maintains stability. A risk factor is the potential dominance of the choleric and the suppression of the phlegmatic's emotional expression.

It is also interesting to examine specific examples of the interaction between cognitive compatibility and temperaments, where radically different types form a unified "decision-making system." In particular:

Steve Jobs (choleric) and Steve Wozniak (melancholic) represent a classic example of collaboration between "vision and detailed technical execution," which played a significant role in the formation of Apple Inc. Jobs, as a leader with a choleric temperament, was primarily oriented toward results, ambitious goals, and market dominance strategies. Wozniak, representing a melancholic temperament, was characterized by precision, attention to detail, and a pursuit of technical perfection. In the decision-making process, Jobs often made fast and high-risk decisions (for example, the Macintosh project), while Wozniak ensured the technical realization of these ideas. The professional tension between them ultimately contributed to the creation of innovative products. [5]

Winston Churchill (choleric) and Franklin D. Roosevelt (sanguine). During World War II, their cooperation represented an important example of complementary leadership temperaments. Churchill was distinguished by strong willpower, energy, and a combative style, which are characteristic of the **choleric temperament**. Roosevelt, on the other hand, is considered a **sanguine-type leader**, with a communicative, optimistic, and diplomatic approach. While Churchill often demanded decisive military

actions, Roosevelt was able to maintain political balance. [6]

Bill Gates (melancholic) and Paul Allen (sanguine) — their collaboration during the early stage of Microsoft's development represents an effective synthesis of temperamental diversity. Gates was characterized by analytical and critical thinking, which is typical of the **melancholic temperament**. He paid particular attention to the detailed review of software code and the identification of potential errors. Allen, representing a **sanguine temperament**, was more focused on generating new ideas, exploring technological opportunities, and developing a broad vision. In the decision-making process, Allen often proposed strategic directions and innovative ideas, while Gates evaluated them through strict logical analysis and technical assessment. [7]

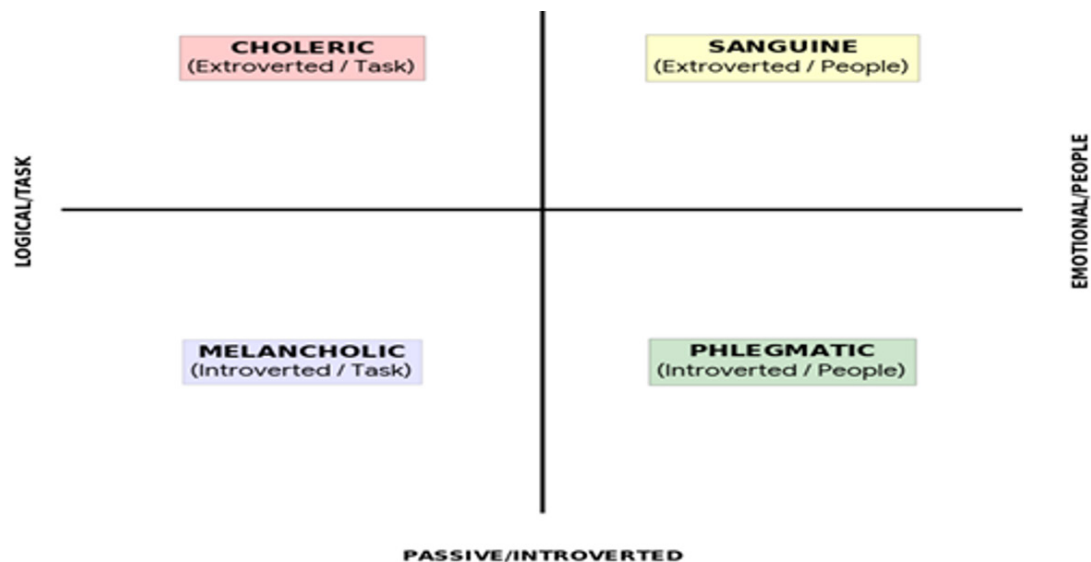
Dwight D. Eisenhower (phlegmatic). During World War II, he stood out as a leader with a **phlegmatic temperament**, performing an important mediating role among military commanders with different personalities and leadership styles. He successfully balanced the positions of strong, **choleric generals** such as G. Patton and B. Montgomery. Eisenhower was distinguished by calmness, patience, and attentive listening. In decision-making, he did not rely on impulsive reactions and instead made balanced, strategically sound decisions. [8]

The “golden rule” of temperaments in practice has been established:

As a rule, the **choleric** initiates the process (energy), the **sanguine** disseminates the idea (communication), the **melancholic** refines the details, and the phlegmatic ensures that the team completes the work in a calm and stable environment (stability). [9]

However, for the purposes of this study, it is important to examine which combinations of individuals with different temperamental styles lead to optimal decision-making outcomes. For this purpose, a cognitive compatibility matrix (team dynamics) is presented. The following figure visually illustrates the distribution of temperaments along two main cognitive axes:

Fig. 1. Cognitive Compatibility Matrix (Team Dynamics)



Source: compiled by the author

Vertical axis (Activity): the upper part includes extraverted, active types (choleric and sanguine), while the lower part includes introverted, relatively passive types (melancholic and phlegmatic).

Horizontal axis (Focus): the left side represents task-oriented and logic-oriented types, while the

right side represents emotion- and people-oriented types.

Analysis of the quadrants in decision-making:

Choleric (active / logical): makes fast, goal-oriented decisions. Motto: "Do it now!"

Sanguine (active / emotional): makes decisions with enthusiasm and intuition, focusing on new opportunities and social influence.

Melancholic (passive / logical): requires deep analysis and data before making decisions. Oriented toward quality and precision.

Phlegmatic (passive / emotional): makes decisions calmly, taking into account team harmony and stability.

The diagram above (Figure 1) represents an ideal cognitive balance. In the decision-making process, the participation of all four types creates a so-called "self-regulating system," where each type compensates for the weaknesses of the others. The choleric provides speed, the melancholic ensures accuracy, the sanguine adds creativity, and the phlegmatic maintains stability. If any quadrant is missing, the decision will either become superficial (in the absence of the melancholic), delayed (in the absence of the choleric), conflictual (in the absence of the phlegmatic), or dull and monotonous (in the absence of the sanguine). [10]

Let us separately consider crisis situations. Organizational crisis management is most effective when individual cognitive differences are integrated in a way that balances rapid decision-making (**choleric**) with thorough risk assessment (**melancholic**). Stable introverts (**phlegmatic**) exhibit a low level of emotional arousal under stress, which enables them to maintain a stable performance level when others experience emotional fluctuations. Studies show that under stress, individuals tend to exaggerate their natural traits. During a crisis:

1. **Choleric: "Crisis Manager"**

Crisis situations are a natural environment for them. They immediately shift into action and take responsibility. Risk: under pressure, the choleric becomes authoritarian and impatient. They may make excessively risky decisions.

2. **Melancholic: "Risk Detector"**

In stressful situations, the melancholic becomes even more focused on data analysis in order to find a safe solution. Risk: "analysis paralysis." In a crisis, when every second matters, the melancholic may become stuck in details and fail to make timely decisions. Individuals with high neuroticism (a trait often associated with melancholic profiles) frequently show performance deterioration under social pressure (so-called "choking under pressure").

3. **Sanguine: "Source of Optimism"**

During crises, the sanguine maintains a positive attitude and looks for creative, non-standard solutions. However, they may underestimate or ignore the level of danger (excessive optimism). Optimistic individuals are generally more resilient to stressors.

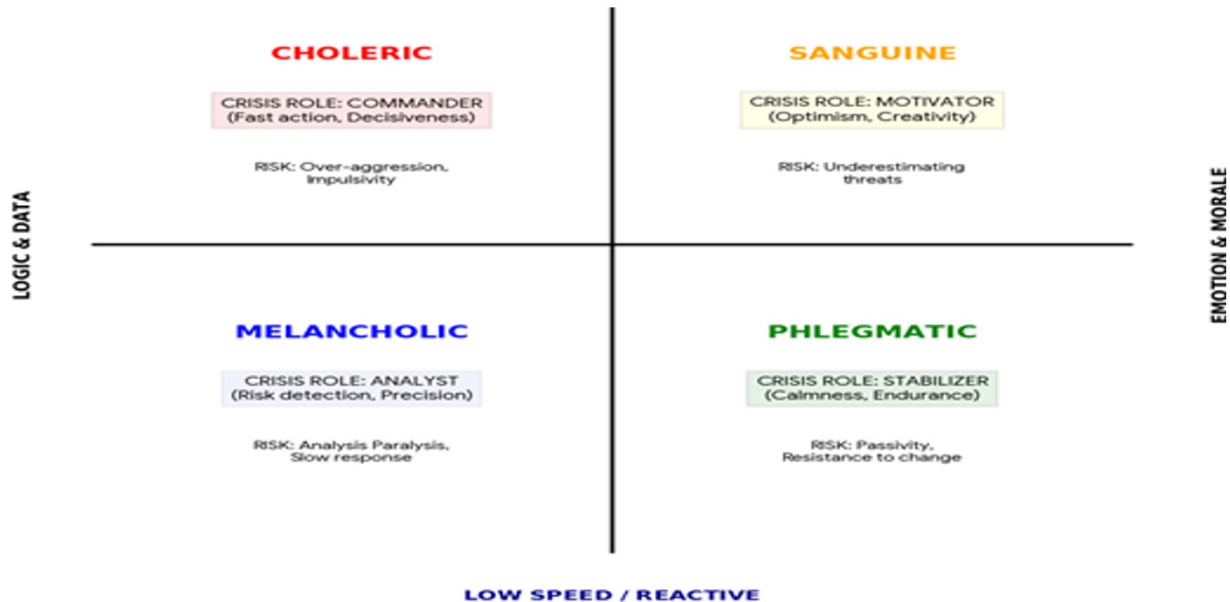
4. **Phlegmatic: "Emotional Stabilizer"**

They maintain remarkable calmness, which positively influences the team during panic situations. Risk: passivity and strong resistance to change. They may struggle to adapt quickly to unexpected changes. In crisis management teams, emotional stability is considered a key indicator of successful performance.

Decisions made under conditions of information scarcity are almost always associated with risk. In such situations, the decision-maker is forced to consider compromise options. For example, in a business environment, a company director who must decide during an economic crisis whether to reduce staff or continue investment faces a high level of risk. Incomplete information increases stress. The final choice is determined by their experience, emotional stability, and attitude toward risk. **Decision-making** is often accompanied by a negative emotional background. Fear of taking responsibility and making

mistakes increases psychological pressure. The larger the scale of the problem, the greater the potential consequences and, accordingly, the higher the level of stress. In extreme cases, this may escalate into panic, which blocks rational thinking. [11]

Fig. 2. Crisis Decision-Making Model



Source: compiled by the author

Let us consider a specific strategy (Figure 2): how to “reconfigure” a team during a crisis in order to avoid a melancholic “paralysis” or excessive risk-taking by the choleric. In crisis situations, the diagram transforms into a functional model in which each type has a specific “battle” role.

Choleric (Commander): Their main function is rapid action. In a crisis, they take the initial impact, but risk becoming overly impulsive.

Sanguine (Motivator): Their role is to maintain morale. They search for creative solutions where others see only dead ends, but may fail to accurately assess real danger.

Melancholic (Analyst): They act as a safety filter. Their analysis is essential to prevent the choleric’s speed from leading to catastrophic outcomes; however, under time pressure they may become “stuck” in analysis.

Phlegmatic (Stabilizer): They are the team’s anchor. Their calmness is an effective remedy for panic, although they may require external stimulation to adapt quickly to rapid changes.

In crisis situations, the greatest risk is the conflict between the choleric and the melancholic (speed vs. precision), which is often balanced by the phlegmatic. [12]

Emotions also play an important role in the decision-making process. A positive emotional state increases cognitive flexibility, supports creative thinking, enhances the ability to generate alternatives, and improves the quality of final decisions. Conversely, stress, anxiety, or fatigue reduce attention, logical analysis, and decision accuracy, thereby increasing the likelihood of errors. The modern digital environment, which includes elements such as artificial intelligence and decision-support systems, places particular emphasis on cognitive compatibility.

CONCLUSION

Cognitive compatibility in the decision-making process represents an important factor that determines the correctness, speed, and effectiveness of decisions. Key influencing factors include the format of information presentation, the level of cognitive load, individual differences, and emotional state. The decision-making process is not only the result of individual intelligence, but also of the temperamental synergy and collaboration among participants in the decision-making process. An ideal team is a unity of all four quadrants, where logic is balanced by emotion, and activity is balanced by observation. This has a direct impact on the decision-making process. Considering cognitive compatibility is essential in any field.

This article particularly highlights the importance of cognitive compatibility among participants in the decision-making process in team work, both in stable and crisis situations.

Ultimately, the more the environment and information correspond to an individual's cognitive capacities, the more high-quality and well-reasoned the resulting decisions become. Overall, cognitive compatibility in human decision-making determines the correctness, speed, and effectiveness of decisions.

Thus, the article presents models of compatibility and collaboration between different temperamental styles in the decision-making process, which can be implemented in commercial and non-commercial organizations, as well as in expert groups or educational institutions.

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ადამიანების კოგნიტური თავსებადობა და კოლაბორაცია გადაწყვეტილების მიღების პროცესში

ზაზა წვერავა

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რეზიუმე

ადამიანის ყოველდღიური საქმიანობა მუდმივად დაკავშირებულია გადაწყვეტილებების მიღებასთან, რომლებიც შეიძლება იყოს როგორც მარტივი, ისე რთული და სტრატეგიული, მაგალითად, ცხოვრების დაგეგმვა, ფინანსური ინვესტიციების განთავსება, სამსახურში სხვადასხვა დონის გადაწყვეტილებების მიღება. გადაწყვეტილების მიღების ეფექტიანობა დამოკიდებულია არა მხოლოდ ხელმისაწვდომ ინფორმაციაზე, არამედ იმაზე, თუ როგორ შეუძლია ადამიანს მისი ალქმა, დამუშავება და შეფასება. ამ პროცესში ერთ-ერთი ძირითადი ფაქტორი არის ადამიანის **კოგნიტურ სტილთან იდენტიფიცირება და სტილებს შორის თავსებადობის გამოვლენა**.

კოგნიტური თავსებადობა ნიშნავს ადამიანის გონებრივ შესაძლებლობებსა და მიღებულ ინფორმაციას შორის შესაბამისობას. ეს მოიცავს ისეთ პროცესებს, როგორიცაა ალქმა, მეხსიერება, ყურადღება, ინტუიცია და ანალიზის უნარი. როდესაც ინფორმაცია შეესაბამება ადამიანის კოგნიტურ სტილს, გადაწყვეტილების მიღება უფრო სწრაფად, ზუსტად და ნაკლები ძალისხმევით მიმდინარეობს. ხოლო თუ ინფორმაცია არათანმიმდევრულია ან რთულად აღსაქმელი, იზრდება შეცდომებისა და არასწორი გადაწყვეტილებების მიღების რისკი.

გადაწყვეტილების მიმღები ადამიანის ინდივიდუალური შესაძლებლობები განსაკუთრებით მნიშვნელოვანია მაშინ, როდესაც გადაწყვეტილება მიიღება კრიზისის, სტრესისა და ინფორმაციული გაურკვევლობის პირობებში. ასეთ სიტუაციებში გადამწყვეტი მნიშვნელობა აქვს არა მხოლოდ პროფესიულ ცოდნასა და გამოცდილებას, არამედ პიროვნების ფსიქოლოგიურ თავისებურებებს, ინტელექტუალურ პოტენციალს, მოტივებს, ღირებულებებსა და შინაგან დამოკიდებულებებს.

სტატიაში წარმოდგენილია და ხაზგასმულია იდეალური გადაწყვეტილების მიღებისათვის, გადაწყვეტილების მიღების პროცესში მონაწილე პირების კოგნიტური თავსებადობის სტილები, ძირითადი მახასიათებლები და მათი კოგნიტური თავსებადობის მატრიცები, როგორც სტაბილური ასევე ცვალებად და კრიზისული სიტუაციების დროს.

სტატიაში ჩამოყალიბებულია სტილებს შორის ტემპერამენტების „ოქროს წესი“ და კოგნიტური თავსებადობის მატრიცები.

საკვანძო სიტყვები: გადაწყვეტილების მიღება, კოგნიტური თავსებადობა,

კოგნიტური ბალანსი, კრიზისი, კოლაბორაცია, „ოქროს წესი“, კოგნიტური თავსებადობის მატრიცა, გადაწყვეტილების ეფექტიანობა, ემოციური მდგომარეობა.